ClassGuard

App Guide & Project Status

Version 2.0: what ClassGuard does, how to install and use it, how each part was tested, and what still has to happen before it goes live.

Hikvision · Dahua · UNV · CP Plus · ONVIF · Any RTSP camera · NVR / DVR · Recording & playback · PTZ · Face attendance · Expression monitor · Safety watch · User roles · Backups

Version	Code	Branch	Date
2.0.0	amit5042k/class-gaurd	claude/camera-management-desktop-app-bR07t	November 2026

1 · AT A GLANCE

What is ClassGuard?

A Windows desktop app (also runs on macOS and Linux), similar to Hikvision iVMS-4200. It shows cameras and NVRs from several brands on one screen, records them, and adds face-recognition attendance, emotion monitoring and safety alerts.

Screens	Camera brands	Automated tests	Planned features built
13	6	71	All

Features

Live View
1×1 to 4×4 grids, pages, groups, single view, full screen, PTZ controls, snapshots, auto-reconnect.

Cameras & NVRs
6 brands, ONVIF scan, stream address fetched from the camera, NVR channel import.

Recording
Continuous, on motion or manual. Day timeline, built-in player, automatic clean-up.

Faces & Attendance
Enroll several photos per person; automatic check-in, Present / Late / Absent, monthly CSV.

Expressions
Live emotion labels and chart; alerts on angry, fear or disgust.

Safety Watch
People in restricted zones; helmet / mask / vest checks with your own YOLO model.

Alerts
Pop-up and sound, snapshot photo, jump straight to the recorded video.

Users & Security
Sign-in, 3 roles, lock screen, audit log, encrypted camera passwords.

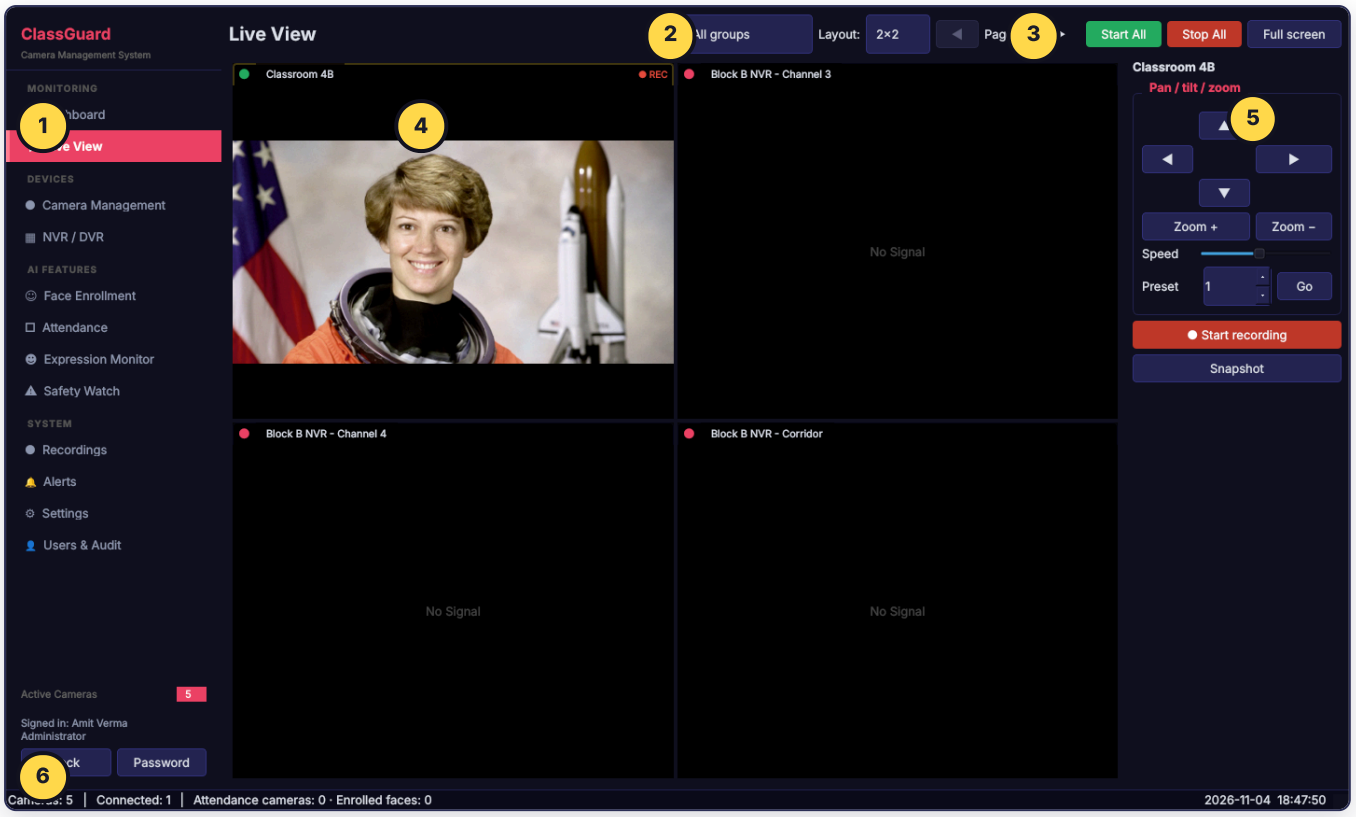
Backup & Install
Daily automatic backups, one-click restore, Windows installer, built-in self-test.

Before going live: everything above has been built and tested against simulated cameras, but **not yet against real Hikvision, Dahua, UNV or CP Plus hardware**. Section 7 lists exactly what still needs checking.

2 · APP TOUR

Finding your way around

A real screenshot of Live View. The menu only shows the screens your account may use.



1. **Menu.** Grouped into Monitoring, Devices, AI features and System.
2. **Group, layout and pages.** Show one group of cameras; ◀ ▶ move between pages when there are more cameras than tiles.
3. **Start / Stop / Full screen.** F11 also switches full screen; Esc leaves it.
4. **Camera tile.** Green dot = connected, ● REC = recording. Click to select, double-click for single view, right-click for snapshot or recording.
5. **Control panel.** Pan / tilt / zoom and presets for PTZ cameras, start or stop recording, snapshot.
6. **Your account.** Shows who is signed in. **Lock** hides the screen without stopping cameras; **Password** changes yours.

All 13 screens

Screen	What it is for	Who can use it
Dashboard	Totals: cameras, connected, people, alerts, attendance, violations	Everyone
Live View	Watch, PTZ, record, snapshot	Everyone*
Attendance · Alerts · Recordings	Daily / monthly attendance; alert list; playback	Everyone*
Face Enrollment · Expression · Safety	People and face photos; emotion and safety monitoring, zones	Operator, Admin
Cameras · NVR / DVR · Settings · Users & Audit	Devices, all settings and backups, accounts and the audit log	Admin

* Viewers can watch and play back but can't move cameras, record, acknowledge alerts or export video.

3 · HOW IT WORKS

Inside the app

Each camera has its own connection thread. Its frames go to three places at once, so a slow AI model or a busy disk never freezes the picture.

Camera / NVR RTSP video over TCP	→ **Stream thread** one per camera; timeouts; reconnects	→ **Screen** up to 15 pictures / s per tile
Stream thread every frame	→ **AI worker** faces 1 / s, safety 2 / s, emotions 1 / s, newest frame only	→ **Attendance, alerts, logs** saved to the database
Stream thread every frame	→ **Recorder** one per camera; motion detection; 10-min MP4 files	→ **Disk clean-up** age, size, free space

Cameras that record or use an AI feature connect automatically when the app starts, like an NVR, even if nobody opens Live View.

Building blocks

Screens	`main_window` `login` `live_view` `recordings` `attendance` `safety_watch` `users` `…13 pages`
Engines	`stream_manager` `recording` `ai_worker` `face_engine` `attendance_engine` `safety_engine` `alerts` `security` `backup`
Brand adapters	`hikvision (ISAPI)` `dahua / CP Plus (CGI)` `onvif` `ptz` `rtsp probe`
Storage	`classguard.db` `settings.json` `secret.key` `faces/` `recordings/` `snapshots/` `backups/`

Where data is kept

All on the PC, in `C:\Users\<you>\.classguard\`. Uninstalling keeps it. Camera passwords inside the database are encrypted with `secret.key`; app sign-in passwords are stored only as one-way hashes.

Install and add cameras

Install on Windows

1. Run **ClassGuard-Setup-2.0.0.exe**. Tick *Start ClassGuard when I sign in* for 24×7 recording.
2. Open ClassGuard and create the **administrator** account (password: 8+ characters, letters and numbers).
3. Optional: Start menu → **ClassGuard self-test** checks every component.
4. In **Users & Audit**, add accounts for staff: *Viewer* for guards, *Operator* for teachers or supervisors.

> The installer includes face recognition. Emotion analysis adds about 1 GB and is installed separately (see the README).

Add cameras

1. **NVR / DVR → + Add**, *Test connection*, then **Import channels**: one camera per channel, named from the recorder.
2. Or **Camera Management → + Add Camera**: brand, IP, login, channel. *Get from camera (ONVIF)* fills in the address for ONVIF cameras.
3. Choose AI features and **Recording: Off / Continuous / Motion**.
4. For PTZ cameras set **Type: PTZ Camera**.

Stream addresses built for each brand

Brand	Main stream (sub-stream)
Hikvision	`rtsp://user:pass@IP:554/Streaming/Channels/101 (102)`
Dahua, CP Plus	`rtsp://user:pass@IP:554/cam/realmonitor?channel=1&subtype=0 (1)`
UNV	`rtsp://user:pass@IP:554/unicast/c1/s0/live (s1)`
ONVIF / other	Fetched from the camera, or paste the camera's own address

If a camera won't connect

What you see	What to check
"Could not open the stream"	IP reachable (ping), port 554 open, RTSP enabled on the camera, correct login
Connects, then drops	Weak Wi-Fi or busy network: tick *Use sub-stream*
NVR test says wrong password	The HTTP port and an account allowed to use the web API
PTZ buttons do nothing	Type is *PTZ Camera*; HTTP port correct; for ONVIF, ONVIF is enabled on the camera

5 · AI FEATURES

Attendance, expressions, safety

1 · Add person
name, ID, class

→

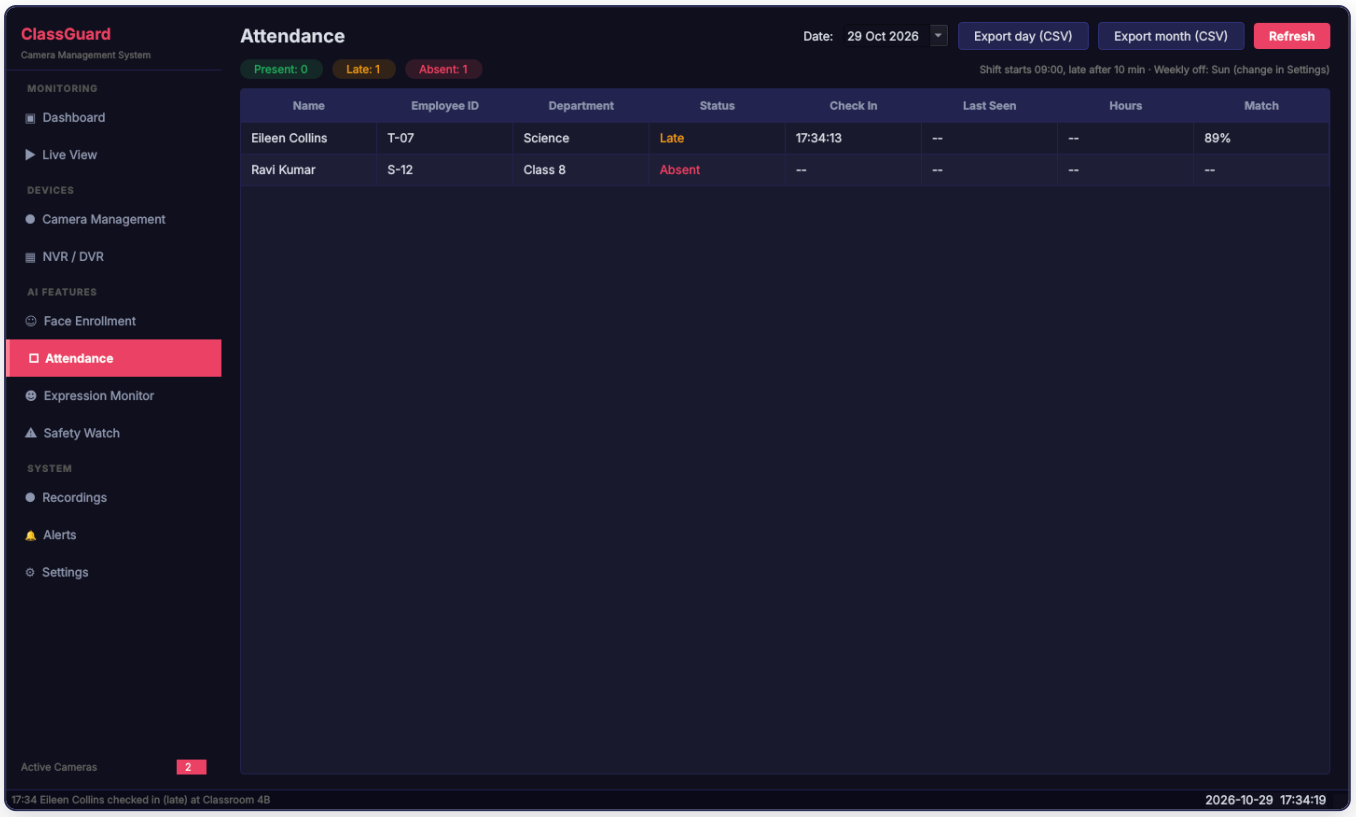
2 · Enroll 3–5 photos
webcam or file

→

3 · Camera sees them
Attendance Tracking on

→

4 · Check-in saved
with photo; last seen updates



- **Late** = first seen after shift start + grace minutes; **Absent** = not seen on a working day. Set both, and weekly off days, in Settings.
- **Monthly CSV**: one row per person with P / L / A / O for every day, plus totals and hours.
- **Match threshold** (Settings, default 0.50): lower means fewer wrong matches but more "Unknown".

Expression Monitor

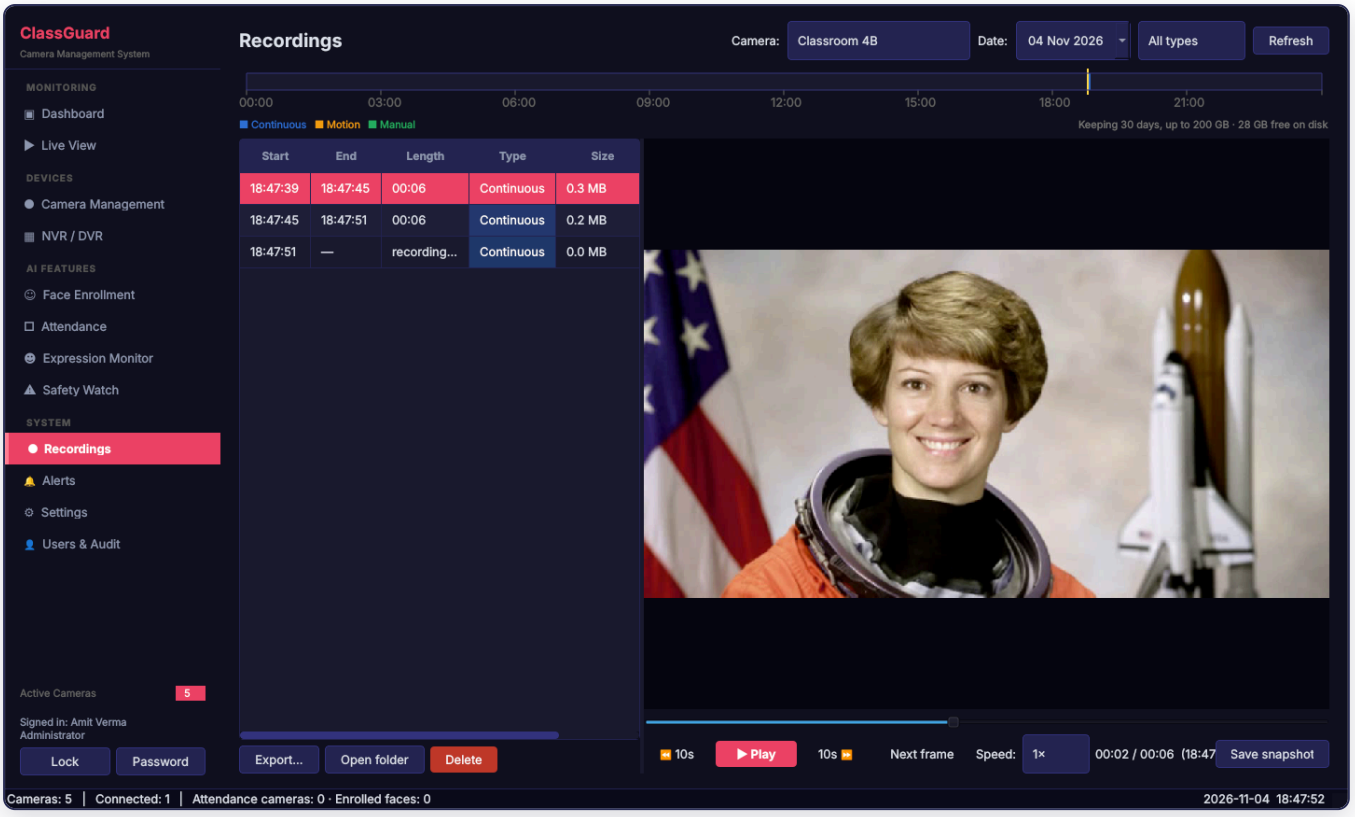
Pick a camera and press Start. Faces are labelled with one of 7 emotions; the bars show how often each appeared. Emotions are also logged every 10 s on cameras with *Expression Monitor* switched on, and angry / fear / disgust above the confidence setting raise an alert. Needs the optional DeepFace install.

Safety Watch

Press **Draw zone**, click round the area, **Save zone**. Anyone whose feet are inside raises a high-priority alert with a photo. For helmet / mask / vest checks, load a YOLO model trained on PPE (`.onnx`, or Darknet `.weights` + `.cfg`) with its class-names file. No PPE model is included.

Privacy: face data is biometric. Tell the people being recorded, get any consent your local law requires, and give guards Viewer accounts.

Recording and playback



- **Timeline**: blue = continuous, orange = motion, green = manual. Click anywhere on it to play from that moment.
- **Player**: 0.5× to 8× speed, ±10 s, frame by frame, snapshot, export to an MP4 file.
- **From an alert**: the **Video** button opens the recording 5 seconds before the alert.
- **Clean-up** (Settings): keep N days, use at most N GB, always leave N GB free. The oldest files go first.

Users and the audit log

Three roles: Viewer, Operator, Administrator (see page 3). After 5 wrong passwords an account is blocked for a minute. There must always be at least one administrator. The audit log records sign-ins (including failed ones), camera, NVR, person and user changes, settings, alert acknowledgements, recordings exported or deleted, and backups.



Backups

A daily automatic backup (last 7 kept), plus **Back up now** and **Restore** in Settings. A backup holds the database, settings, face photos and the encryption key, but not recordings. Restores are applied on restart, and the previous data is kept in a separate folder.

7 · PROJECT STATUS

What's built and how it was checked

SIMULATED CAMERA tested end-to-end with a simulated RTSP camera AUTOMATED TEST covered by automated tests NEEDS HARDWARE must be confirmed on real devices

Area	Status	Verification
Streaming, reconnect, grid	Built	SIMULATED CAMERA NEEDS HARDWARE
Brand stream addresses	Built	AUTOMATED TEST NEEDS HARDWARE
Recording (3 modes), clean-up	Built	SIMULATED CAMERA AUTOMATED TEST
Playback, timeline, alert → video	Built	AUTOMATED TEST
Face enrollment & recognition	Built	SIMULATED CAMERA (real face photo)
Attendance & reports	Built	SIMULATED CAMERA AUTOMATED TEST
Alerts, zones, pop-ups	Built	AUTOMATED TEST
YOLO model loading (ONNX / Darknet)	Built	AUTOMATED TEST (format only, no real PPE model)
Emotion analysis	Built	NEEDS HARDWARE model couldn't be downloaded in the build environment
PTZ (Hikvision, Dahua, ONVIF)	Built	AUTOMATED TEST (commands sent) NEEDS HARDWARE
NVR test & channel import	Built	AUTOMATED TEST NEEDS HARDWARE
ONVIF discovery & address fetch	Built	NEEDS HARDWARE
Users, roles, lock, audit	Built	AUTOMATED TEST
Encrypted passwords + upgrade	Built	AUTOMATED TEST
Backup / restore	Built	AUTOMATED TEST
Packaged app	Built	Built and self-tested on Linux; Windows build set up in GitHub Actions

Known limitations

- Up to about 16 cameras for viewing and recording on a normal PC. AI on more than 4–6 cameras at once needs a stronger PC; there is no GPU acceleration.
- Recordings are made by ClassGuard itself. Playing back footage stored *on* the NVR's own disk is not supported yet.
- Recordings are MP4 (MPEG-4 Part 2). They play in ClassGuard and VLC; some other players may not open them.
- No mobile app or web access; no email or WhatsApp alerts.

8 · NEXT STEPS

Going live

Step 1
Hardware trial
1 week

- Install on the target PC and run the self-test.
- Connect one camera of each brand plus one NVR, wired and Wi-Fi, and work through the checklist below.
- Fix any stream-address or PTZ differences found (most can be handled with *Custom RTSP URL*).

Step 2
Pilot
2 weeks

- Run 24×7 in one building: recording, attendance in one classroom, one safety zone.
- Tune the match threshold, motion sensitivity and disk limits; check daily backups are being made.

Step 3
Optional extras

- Play back footage stored on the NVR; GPU acceleration; email / WhatsApp alerts; a mobile viewer.
- A trained helmet / vest model for workshops and labs.

Hardware acceptance checklist

- [ ] Hikvision, Dahua, UNV and CP Plus cameras stream
- [ ] NVR: test shows model; Import channels creates cameras
- [ ] ONVIF Discover finds a camera; address fetched
- [ ] Wi-Fi camera recovers after a router restart
- [ ] PTZ moves, stops when released, goes to a preset
- [ ] 16 cameras in 4×4 for 1 hour without slowdown
- [ ] Continuous and motion recordings play back
- [ ] Disk limit deletes the oldest recordings
- [ ] Enrolled person recognised at 1–3 m
- [ ] Check-in, Late and Absent correct; CSV opens in Excel
- [ ] Zone intrusion raises an alert with photo and video
- [ ] Viewer account can't change anything
- [ ] Backup restored on a second PC; cameras reconnect
- [ ] App restarts by itself after a Windows reboot

Summary. Every planned feature is built, and 71 automated tests pass, including complete runs against a simulated camera. What's left is to prove it on your real cameras and NVRs (Step 1) and then run a short pilot.